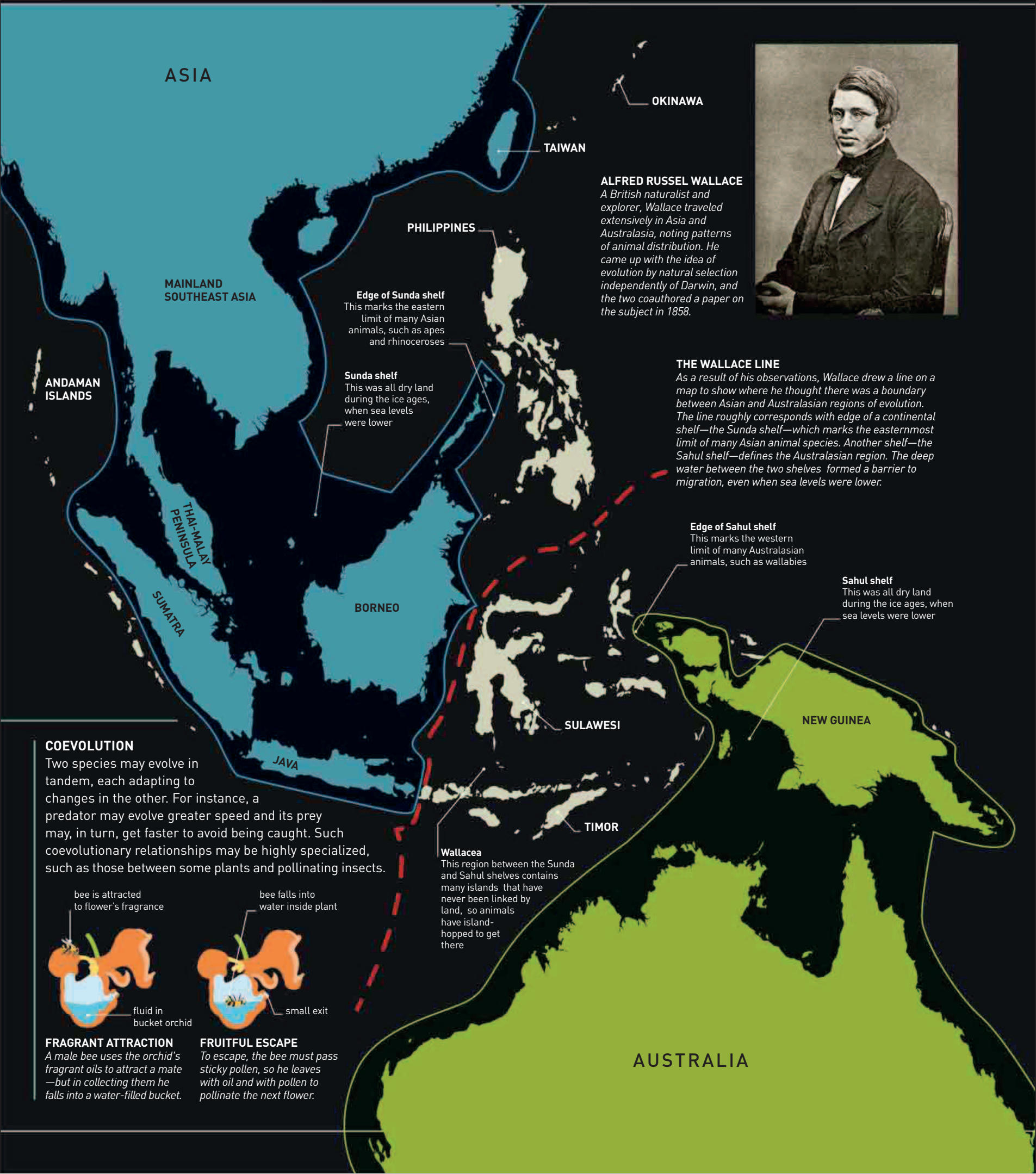




ASIA

OKINAWA

TAIWAN

PHILIPPINES

MAINLAND
SOUTHEAST ASIA

Edge of Sunda shelf
This marks the eastern limit of many Asian animals, such as apes and rhinoceroses

Sunda shelf
This was all dry land during the ice ages, when sea levels were lower

ANDAMAN ISLANDS

THAI-MALAY
PENINSULA

SUMATRA

BORNEO

JAVA

SULAWESI

TIMOR

NEW GUINEA

AUSTRALIA

ALFRED RUSSEL WALLACE

A British naturalist and explorer, Wallace traveled extensively in Asia and Australasia, noting patterns of animal distribution. He came up with the idea of evolution by natural selection independently of Darwin, and the two coauthored a paper on the subject in 1858.



THE WALLACE LINE

As a result of his observations, Wallace drew a line on a map to show where he thought there was a boundary between Asian and Australasian regions of evolution. The line roughly corresponds with edge of a continental shelf—the Sunda shelf—which marks the easternmost limit of many Asian animal species. Another shelf—the Sahul shelf—defines the Australasian region. The deep water between the two shelves formed a barrier to migration, even when sea levels were lower.

Edge of Sahul shelf

This marks the western limit of many Australasian animals, such as wallabies

Sahul shelf

This was all dry land during the ice ages, when sea levels were lower

Wallacea

This region between the Sunda and Sahul shelves contains many islands that have never been linked by land, so animals have island-hopped to get there

COEVOLUTION

Two species may evolve in tandem, each adapting to changes in the other. For instance, a predator may evolve greater speed and its prey may, in turn, get faster to avoid being caught. Such coevolutionary relationships may be highly specialized, such as those between some plants and pollinating insects.

bee is attracted to flower's fragrance

bee falls into water inside plant



fluid in bucket orchid



small exit

FRAGRANT ATTRACTION

A male bee uses the orchid's fragrant oils to attract a mate—but in collecting them he falls into a water-filled bucket.

FRUITFUL ESCAPE

To escape, the bee must pass sticky pollen, so he leaves with oil and with pollen to pollinate the next flower.